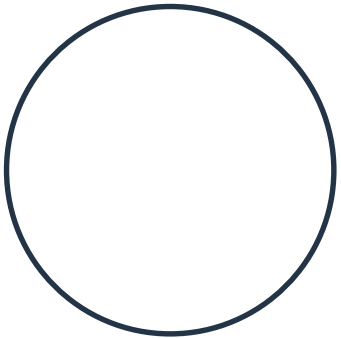
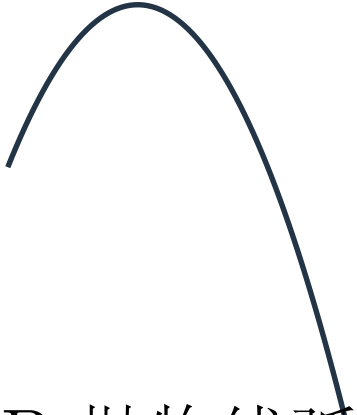
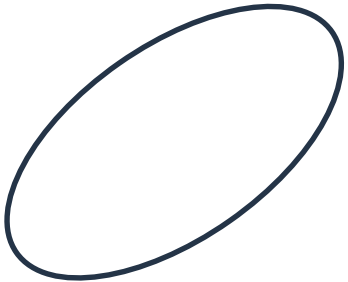


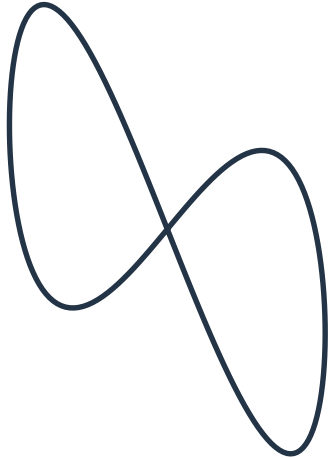
A 圆



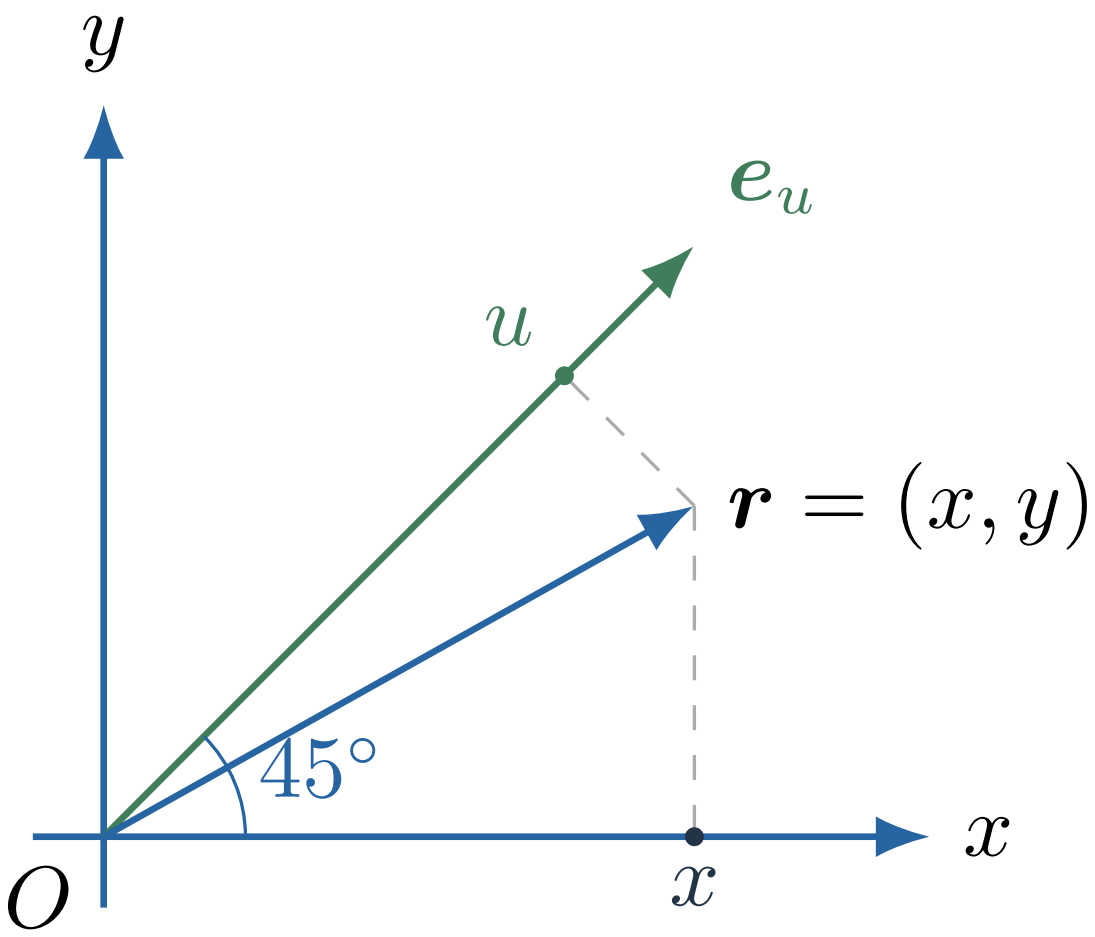
B 抛物线弧



C 椭圆



D 八字形

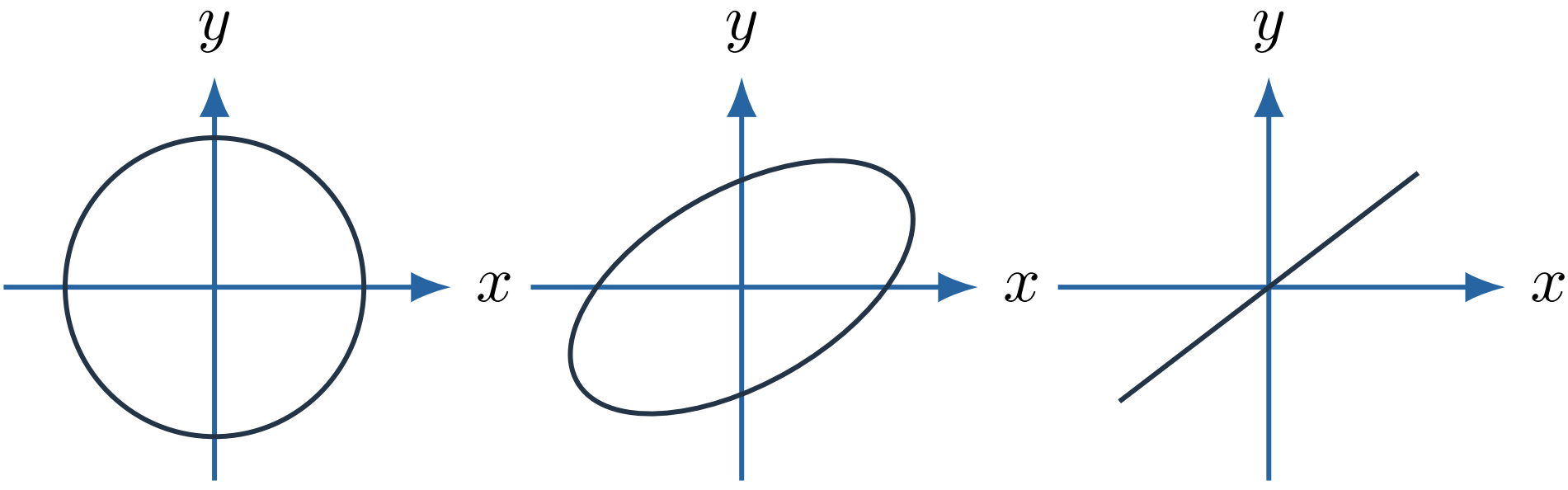


$$u = \boldsymbol{r} \cdot \boldsymbol{e}_u$$

$$u = \frac{x + y}{\sqrt{2}}$$

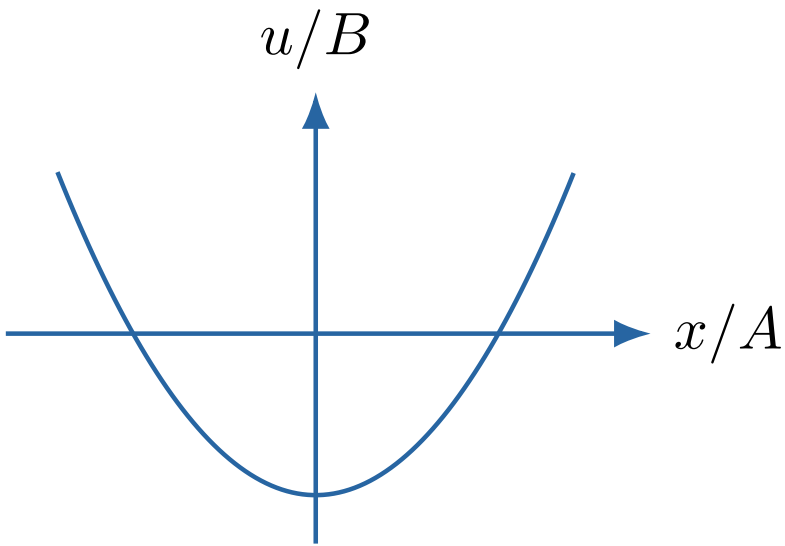
$$y = \sqrt{2}u - x$$

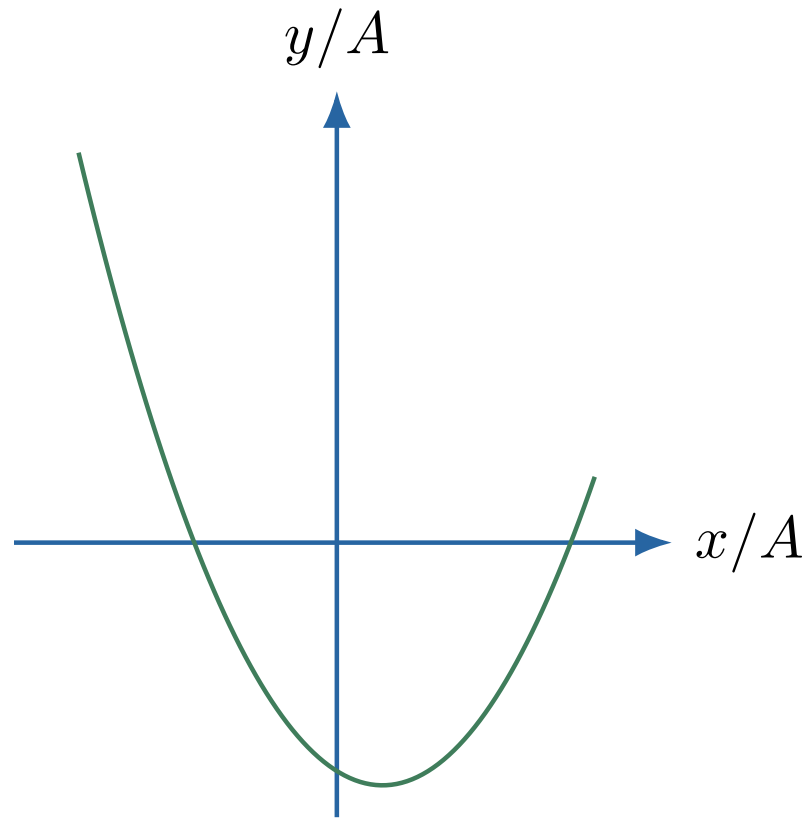
辅助坐标 (x, u) 不是一组真实正交方向



圆　　椭圆　　线段

两个投影同频；相位差为 0 或 π 时可退化

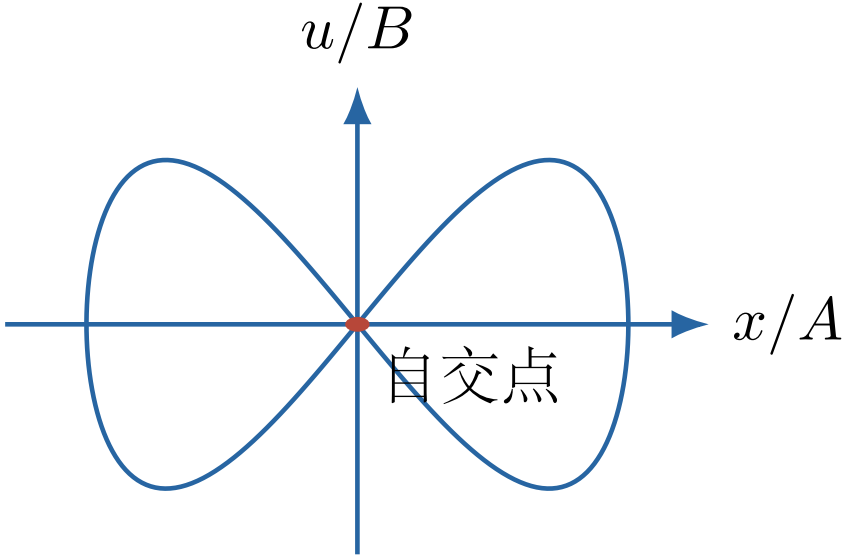




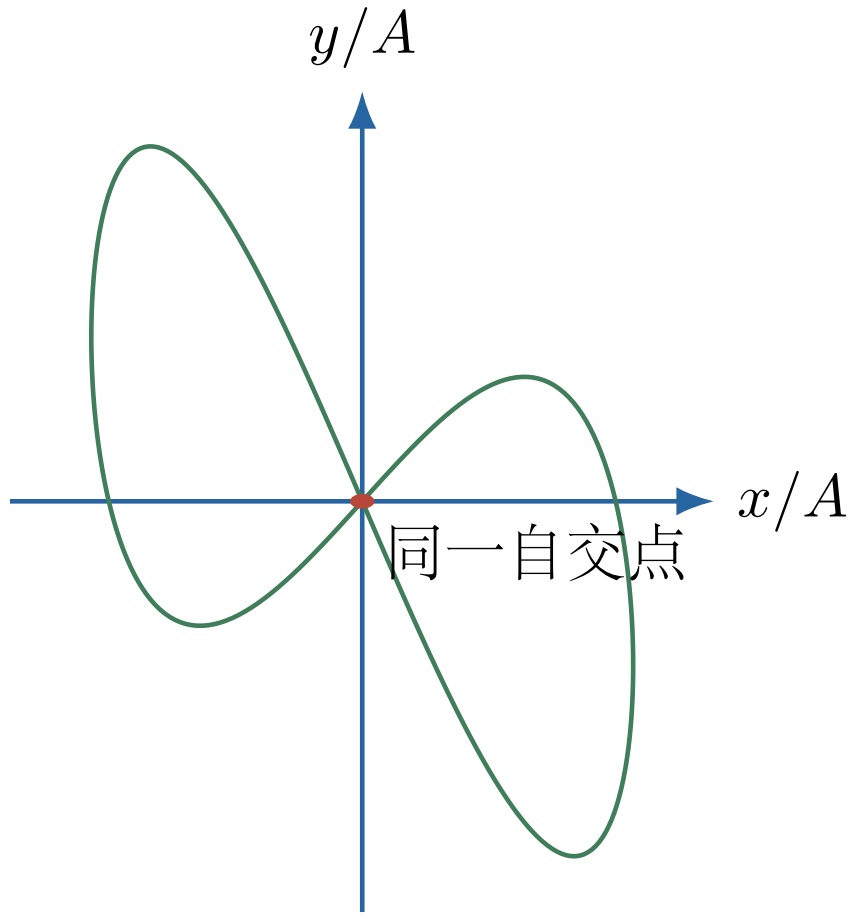
辅助平面：$u/B = 2(x/A)^2 - 1$

真实平面：$A = B$ 的有限弧

$$x = A\cos\omega t, \quad u = B\cos 2\omega t, \quad y = \sqrt{2}u - x$$



辅助八字形



真实八字形，$A = B$

$$x = A\sin\omega t, \quad u = B\sin 2\omega t, \quad y = \sqrt{2}u - x$$